FedXAI: A Privacy-Preserving Explainable Federated Multimodal Learning Framework for Alzheimer's Disease Diagnosis

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***Abstract*—Clinicians rarely diagnose Alzheimer's disease (AD) from a single source. They read a structural brain MRI alongside cognitive test scores, demographic details, and clinical biomarkers. Most deep learning models for AD don't work this way. They use only one of these inputs, or they assume every patient's data can be gathered in one place, which hospitals are often not allowed to do. They also say little about why they made a given prediction. We present FEDXAI, a federated multimodal framework for AD diagnosis that keeps patient data on site and explains its predictions. A 3D Convolutional Neural Network (3D CNN) encodes the MRI volume, a multilayer perceptron (MLP) encodes the clinical variables, and intermediate fusion combines the two feature vectors. Each institution trains on its own data, so raw scans and records never leave the premises. Only model updates are shared, clipped and perturbed with Gaussian noise once per round, then combined with Federated Averaging (FedAvg). For each prediction, 3D Grad-CAM highlights the brain regions involved and SHAP ranks the clinical variables. We describe the framework's design and a protocol for evaluating it on data from the Alzheimer's Disease Neuroimaging Initiative (ADNI) under centralized, IID, and non-IID training. The software is implemented and has run end to end on ADNI clinical data with synthetic placeholder images. We haven't used real MRI scans yet, so we report no accuracy results.**

***Index Terms*—Federated learning, explainable AI, multimodal deep learning, 3D CNN, privacy, medical diagnosis, Grad-CAM, SHAP, non-IID data, FedAvg, Alzheimer's disease.**

I. INTRODUCTION

Deep learning already does well on many medical imaging tasks, and on some radiology and pathology problems its accuracy comes close to that of specialists. Getting these models into everyday clinical practice is harder. A doctor almost never decides from a scan alone. The scan is read together with the patient's history, examination findings, cognitive test results, and risk factors such as age [1]. A model that sees only one of these sources misses useful context, and that can make it less sensitive and less robust [2].

Two more problems make multimodal models hard to build across hospitals. The first is privacy. Laws such as HIPAA in the United States and GDPR in the European Union strictly limit how health records and medical images can be shared, and copying patient data into a central database carries legal and institutional risk [3]. So each hospital often ends up training on its own small cohort, and models trained that way tend to overfit and generalize poorly. The second is trust. A deep network doesn't explain its output, and clinicians can't reasonably act on a probability score when they have no way to check the reasoning behind it [4].

Federated Learning (FL) sidesteps the data-sharing problem. Several institutions train one model together without moving their data [5]. Each site keeps its records locally and sends only model parameters (or gradients) to a coordinating server over an encrypted connection, which in our case is gRPC secured with TLS. In FEDXAI, the server combines these updates with Federated Averaging (FedAvg) [5], weighting each client's parameters by the number of samples it holds. FedAvg means data no longer has to be centralized, but the differences between sites remain. Patient populations, clinical protocols, and scanners vary from one hospital to the next, so each client's data is not independent and identically distributed (non-IID) [6].

Explainable AI (XAI) methods can show what a trained model relies on. Grad-CAM [7] produces heatmaps over the input image, and SHAP [8] assigns an importance value to each input feature. But most earlier work treated FL and XAI separately, or applied XAI only to image-only models. Much less is known about whether the explanations from a multimodal model stay reliable when it's trained across heterogeneous federated clients.

*Data used in preparation of this article were obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database (adni.loni.usc.edu). As such, the investigators within the ADNI contributed to the design and implementation of ADNI and/or provided data but did not participate in analysis or writing of this report. A complete listing of ADNI investigators can be found at: http://adni.loni.usc.edu/wp-content/uploads/how_to_apply/ADNI_Acknowledgement_List.pdf

FEDXAIcombines intermediate multimodal fusion, FedAvg-based federated training, clipped and noise-perturbed updates, and explanations for both the imaging and the clinical inputs. We present its design for Alzheimer's disease and a protocol for evaluating it on paired 3D MRI scans and clinical records, under centralized training and under controlled levels of data skew. The software is implemented, but it hasn't been trained on real MRI scans yet, so we report no accuracy results. The main contributions of this work are:

- A two-branch multimodal architecture. A 4-stage 3D CNN encodes the brain MRI (128-d), an MLP encodes the clinical markers (32-d), and the two outputs are fused into a 160-d representation.
- Federated training with privacy protections, built on PyTorch and Flower over gRPC. Each client's update is clipped and perturbed with Gaussian noise once per round, then combined with FedAvg weighted by sample size.
- Explanations for both modalities: a 3D Grad-CAM heatmap and a set of SHAP feature attributions for each prediction.
- An evaluation protocol that sets out the planned comparisons and measures for centralized training and for IID and Dirichlet non-IID clients, scored on held-out subjects only.

Section II reviews related work and the research gap. Section III describes the architecture and method, and Section IV gives the dataset and settings. Section V covers the implementation status and the evaluation protocol, and Section VI concludes.

II. Related Work

Our work draws on four lines of research.

A. Deep learning for Alzheimer's disease diagnosis

CNNs are a common choice for classifying Alzheimer's disease from structural MRI [9], [10]. Qiu *et al.* [9] combined MRI with routinely collected clinical data to separate normal cognition, mild cognitive impairment, and dementia, and used SHAP to tie the model's decisions to brain regions. A model that sees only the image can't use what's in clinical notes and test scores [11], which is why FEDXAI pairs its 3D CNN with a clinical branch.

B. Multimodal fusion for clinical diagnosis

Fusion methods are usually grouped into early fusion of inputs, late fusion of predictions, and intermediate fusion of learned features [2], [11]. Models that combine imaging with electronic health record data have generally outperformed single-modality models [2], [11], since the two sources carry complementary information [1], [11]. But most are trained on data pooled at one site, which is hard to arrange under current privacy rules [1], [3].

C. Federated learning and privacy protection

FL lets institutions train a model without exchanging patient records [3], [10], [12]. In FedAvg [5], the server averages the client updates, weighted by dataset size. Applications include ADMarker [12], which detects digital biomarkers of Alzheimer's disease from sensor data recorded in natural living environments, and FedVC-ADDiM [10], which diagnoses the disease from MRI scans held at different sites. There is also a benchmark of FL algorithms for medical image classification [13]. Shared parameters can still reveal information about the training data, so differential privacy is often added [3], [14]. Two problems remain. Non-IID client data slows convergence and can lower accuracy [6], [13], and federated medical studies say little about whether the aggregated model is still interpretable [13]. Our protocol therefore evaluates FedAvg under controlled Dirichlet label skew [15].

D. Explainable AI

Clinicians are more likely to adopt a model whose decisions they can inspect [4]. Grad-CAM [7] builds a heatmap from the gradients flowing into the last convolutional layer. For a 3D layer, the result is a volume that can be overlaid on the scan. SHAP [8] uses Shapley values to attribute a prediction to tabular features. Jahan *et al.* [16] used SHAP to explain a federated multimodal AD model. But explanation quality can't be taken for granted. Jin *et al.* [17] tested 16 heatmap methods on a multimodal medical imaging task, and none reliably reflected the model's decision process.

E. Research gap and our approach

Four things are missing from the work above. Image-only 3D CNNs ignore clinical information [9]. Fusion models usually assume centralized data [2], [11]. Federated studies concentrate on accuracy [6], [13]. And federated XAI studies have rarely paired a visual explanation with a tabular one for a single fused model [16], [17]. FEDXAI is built to address all four. It fuses a 3D CNN and an MLP (Sections III-C and III-D), trains the fused model with FedAvg and clipped, noise-perturbed updates (Section III-E), and explains each prediction for both modalities (Section III-F).

III. Proposed FEDXAI framework

A. System architecture overview

Fig. 1 shows the architecture of FEDXAI. It has two layers. The *local layer* runs inside each hospital, and the *federated layer* coordinates training between hospitals. Only model parameters pass between them. MRI scans and clinical records never leave the hospital.

Each hospital holds a structural 3D MRI volume and a clinical record for every subject. A four-stage 3D CNN encodes the preprocessed volume into a 128-d vector, and an MLP encodes the clinical record into a 32-d vector. The two are concatenated, and a small classifier maps the 160-d result to the class probabilities $\hat{y}$. 3D Grad-CAM and SHAP then explain the prediction. In training, each hospital updates its own copy of the model, clips the update, and adds Gaussian

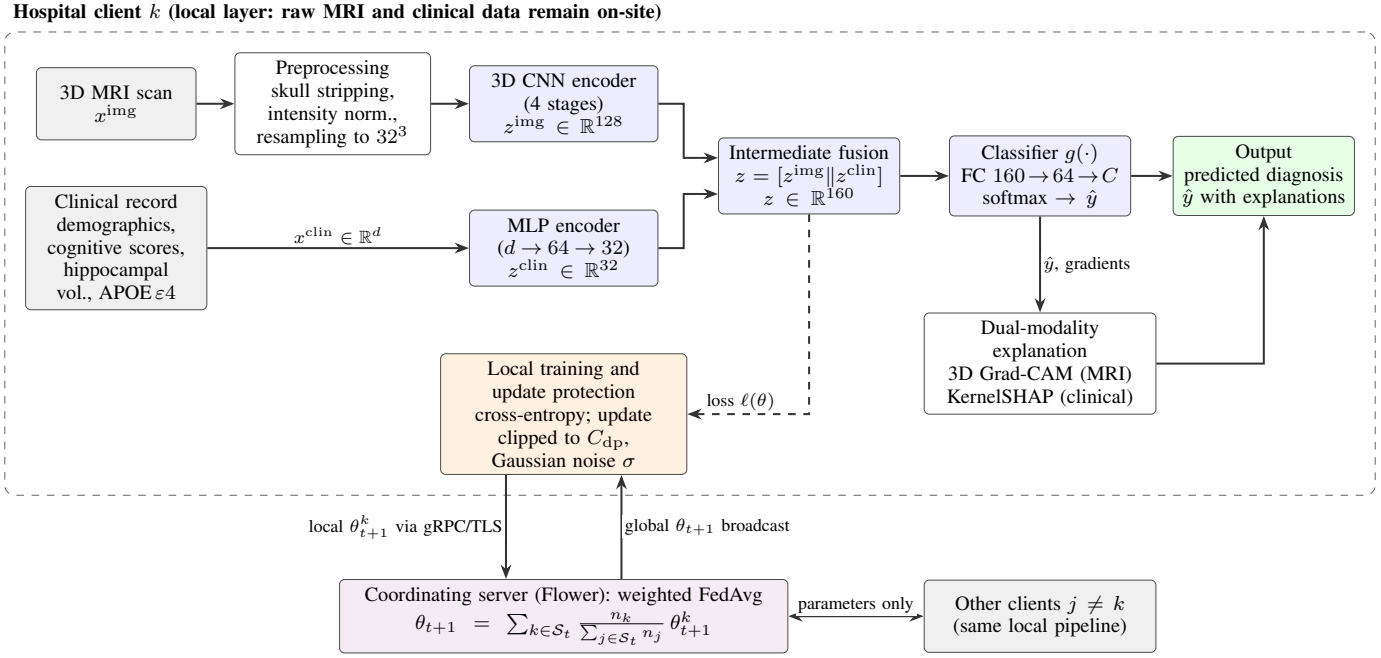


Fig. 1. Architecture of FEDXAI. Inside each hospital client (dashed box), a 3D CNN encodes the preprocessed MRI volume and an MLP encodes the clinical feature vector. The two are fused into a 160-d vector and classified, and 3D Grad-CAM and KernelSHAP explain the prediction. During training, each client sends its clipped, noise-perturbed update over gRPC/TLS (Flower) to the server. The server runs weighted FedAvg and sends back the new global model. Patient data stays at the client.

noise at the end of the round. It then sends the result to the server through Flower over gRPC/TLS, and the server applies weighted FedAvg.

B. Problem formulation

We consider K healthcare institutions, indexed by $k \in \{1, \ldots, K\}$. Institution k holds a local dataset $\mathcal{D}_k = \{(x_i^{\text{img}}, x_i^{\text{clin}}, y_i)\}_{i=1}^{n_k}$, where $x_i^{\text{img}} \in \mathbb{R}^{1 \times D \times H \times W}$ is a 3D MRI volume, $x_i^{\text{clin}} \in \mathbb{R}^d$ is the matching clinical vector, and $y_i \in \{1, \ldots, C\}$ is the diagnostic label. The total number of samples is $n = \sum_{k=1}^{K} n_k$. Each $\mathcal{D}_k$ stays on its own client server during both training and inference.

C. Multimodal encoder architecture

The imaging branch $f_{\text{img}}(\cdot; \theta_{\text{img}})$ is a 3D CNN with four stages. Every stage applies 3D convolution, batch normalization, a LeakyReLU activation (negative slope 0.1), and 3D max pooling. A global average pooling layer then reduces the feature maps to a 128-dimensional vector:

$$z^{\text{img}} = f_{\text{img}}(x^{\text{img}}; \theta_{\text{img}}) \in \mathbb{R}^{128}. \tag{1}$$

The clinical branch $f_{\text{clin}}(\cdot; \theta_{\text{clin}})$ is an MLP. It passes the d clinical features through hidden layers of size 64 and 32 with ReLU activations and dropout ($p = 0.1$), and outputs a 32-dimensional vector:

$$z^{\text{clin}} = f_{\text{clin}}(x^{\text{clin}}; \theta_{\text{clin}}) \in \mathbb{R}^{32}. \tag{2}$$

D. Intermediate feature fusion

We concatenate the two feature vectors into a 160-dimensional joint vector z:

$$z = \left[z^{\text{img}} \parallel z^{\text{clin}} \right] \in \mathbb{R}^{160}. \tag{3}$$

A fully connected classifier $g(\cdot; \theta_c)$ ($160 \to 64 \to C$, Table I) then gives the probability of each of the C classes:

$$\hat{y} = \text{softmax}\big(g(z; \theta_c)\big). \tag{4}$$

Let $\theta = \{\theta_{\text{img}}, \theta_{\text{clin}}, \theta_c\}$ be all trainable parameters. The local model is trained with cross-entropy loss:

$$\ell(\theta) = -\sum_{c=1}^{C} y_c \log \hat{y}_c. \tag{5}$$

Client k's local objective is the average of this loss over its dataset, $F_k(\theta) = \frac{1}{n_k} \sum_{i=1}^{n_k} \ell_i(\theta)$, where ℓ_i is the loss on sample i of $\mathcal{D}_k$.

E. Update protection and FedAvg aggregation

Model updates can leak information about the training data, so each client protects its update before sending it. Let θ_t be the global model received at round t and θ^k_{loc} the model of client k after local training. Client k's update is $\Delta^k = \theta^k_{\text{loc}} - \theta_t$. Once per round, the client clips this whole update so that its L_2 norm is at most C_{dp}:

$$\bar{\Delta}^k = \Delta^k \,/\, \max\left(1, \frac{\|\Delta^k\|_2}{C_{\text{dp}}}\right). \tag{6}$$

It then adds Gaussian noise with multiplier σ and returns the protected model:

$$\theta_{t+1}^{k} = \theta_t + \bar{\Delta}^k + \boldsymbol{\xi}, \qquad \boldsymbol{\xi} \sim \mathcal{N}\big(\mathbf{0},\, \sigma^2 C_{\mathrm{dp}}^2\, \mathbf{I}\big). \tag{7}$$

This differs from DP-SGD [14], which clips the gradient of every sample at every step. Clipping a client's update bounds that client's influence on the global model, not the influence of a single patient record. Also, with $\sigma = 0.1$ the added noise is small, and we carry out no privacy accounting. We therefore make no formal differential privacy claim and state no (ϵ, δ) value. The mechanism is a practical safeguard and carries no certified privacy guarantee.

The privacy of FEDXAI rests on three layers. Raw MRI scans and clinical records stay at the hospital. Model updates travel over TLS, which protects them from outside parties in transit. Clipping and noise limit what can be inferred about a patient from an update. We assume an honest-but-curious server that follows the protocol but may inspect what it receives. Secure aggregation is not used, so the server sees each client's update separately, and clipping and noise are the only protection against it.

The overall goal is to minimize the loss over the patients of all institutions:

$$\min_{\theta}\; F(\theta) = \sum_{k=1}^{K} \frac{n_k}{n}\, F_k(\theta). \tag{8}$$

At round t, the server sends the global weights θ_t to the clients. Each active client in $\mathcal{S}_t \subseteq \{1, \ldots, K\}$ trains locally and returns θ_{t+1}^{k}. The server then averages these weights, weighting each client by its sample size, to get θ_{t+1}:

$$\theta_{t+1} = \sum_{k \in \mathcal{S}_t} \frac{n_k}{\sum_{j \in \mathcal{S}_t} n_j}\, \theta_{t+1}^{k}. \tag{9}$$

Algorithm 1 gives the complete training procedure, including the privacy steps.

Algorithm 1 FEDXAI Privacy-Preserving FedAvg Protocol

Require: Client set $\{1, \ldots, K\}$, rounds R, local epochs E, learning rate η, clipping norm C_{dp}, noise multiplier σ
Ensure: Final global model parameters θ_R

Server initializes θ_0
for $t = 0$ **to** $R - 1$ **do**
  Select active client subset $\mathcal{S}_t \subseteq \{1, \ldots, K\}$
  Broadcast global model θ_t to each client $k \in \mathcal{S}_t$ via gRPC/Flower
  for each client $k \in \mathcal{S}_t$ **in parallel do**
    $\theta_{t+1}^{k} \leftarrow \textsc{ClientUpdate}(k, \theta_t, C_{\mathrm{dp}}, \sigma)$
  end for
  $\theta_{t+1} \leftarrow \sum_{k \in \mathcal{S}_t} \frac{n_k}{\sum_{j \in \mathcal{S}_t} n_j}\, \theta_{t+1}^{k}$ {Weighted FedAvg}
end for
return θ_R

function $\textsc{ClientUpdate}(k, \theta_t, C_{\mathrm{dp}}, \sigma)$
$\theta \leftarrow \theta_t$
for local epoch $= 1$ **to** E **do**
  for all mini-batches $(x^{\mathrm{img}}, x^{\mathrm{clin}}, y) \in \mathcal{D}_k$ **do**
    $z \leftarrow [\, f_{\mathrm{img}}(x^{\mathrm{img}}) \parallel f_{\mathrm{clin}}(x^{\mathrm{clin}})\,]$
    $\hat{y} \leftarrow \mathrm{softmax}(g(z))$
    Compute gradient $\mathbf{v} \leftarrow \nabla_\theta \ell(\hat{y}, y)$
    $\theta \leftarrow \textsc{AdamStep}(\theta, \mathbf{v}, \eta)$
  end for
end for
Update $\Delta \leftarrow \theta - \theta_t$
Clip update $\bar{\Delta} \leftarrow \Delta / \max(1, \|\Delta\|_2 / C_{\mathrm{dp}})$ {Eq. (6)}
return $\theta_t + \bar{\Delta} + \mathcal{N}(\mathbf{0}, \sigma^2 C_{\mathrm{dp}}^2 \mathbf{I})$ {Eq. (7)}

F. Multimodal explanation integration

For the MRI input, 3D Grad-CAM computes a weight w_q^c for each channel q of the last convolutional feature maps A^q, where Z is the number of voxels in a map:

$$w_q^c = \frac{1}{Z} \sum_i \sum_j \sum_l \frac{\partial y^c}{\partial A_{ijl}^q}, \qquad L_{\mathrm{cam}}^c = \mathrm{ReLU!}\Big(\sum_q w_q^c A^q \Big). \tag{10}$$

The heatmap is upsampled to the input resolution and overlaid on the MRI volume. With a 32^3 input and three pooling steps before the final convolution, each A^q has only $4 \times 4 \times 4$ voxels. The heatmap therefore shows coarse brain regions and can't resolve small structures such as the hippocampus.

For the clinical input, KernelSHAP estimates the Shapley value ϕ_i of feature i, its average contribution to the prediction over subsets of the feature set F:

$$\phi_i = \sum_{S \subseteq F \setminus i} \frac{|S|!, (|F| - |S| - 1)!}{|F|!}, \big[f_x(S \cup i) - f_x(S) \big]. \tag{11}$$

G. Data heterogeneity partitioning

To create non-IID clients we use a Dirichlet distribution $\mathrm{Dir}(\alpha)$ with concentration parameter α [15]. For each class c, the share of that class assigned to each of the K clients is drawn as:

$$(p_{c,1}, \ldots, p_{c,K}) \sim \mathrm{Dir}(\alpha \cdot \mathbf{1}_K). \tag{12}$$

Smaller values of α give stronger label skew, much like the gap between general hospitals and specialist clinics.

IV. DATASET AND SETTINGS

A. Dataset and preprocessing

Data used in the preparation of this article were obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database (adni.loni.usc.edu). The ADNI was launched in 2003 as a public-private partnership, led by Principal Investigator Michael W. Weiner, MD. The primary goal of ADNI has been to test whether serial magnetic resonance imaging (MRI), positron emission tomography (PET), other biological markers, and clinical and neuropsychological assessment can be combined to measure the progression of mild cognitive impairment (MCI) and early Alzheimer's disease (AD).

The clinical variables come from the ADNI study tables [18]. They are demographics (age, sex, education), cognitive scores (MMSE, CDR-SB, ADAS-Cog13), hippocampal

TABLE I
HYPERPARAMETERS AND EXPERIMENTAL SETTINGS

Parameter	Configuration
Image Backbone	4-Stage 3D CNN, 128-d Embedding
Clinical Net	MLP ($d \to 64 \to 32$), Dropout 0.1, 32-d Embedding
Feature Fusion	Latent Concatenation (160-d)
Classifier Head	FC ($160 \to 64 \to C$)
Optimization	Adam ($\eta = 10^{-3}$, Weight Decay 10^{-4})
Local Training	Local Epochs $E = 1$, Batch Size 16
Federation Bounds	Communication Rounds $R = 50$, Clients $K = 5$
Update Protection	Client update clipped once per round, $C_{\mathrm{dp}} = 1.0$, $\sigma = 0.1$
Data Skew Bounds	Uniform IID vs. Dirichlet $\mathrm{Dir}(\alpha), \alpha \in \{1.0, 0.5, 0.1\}$
FL Framework	PyTorch + gRPC / Flower
Aggregator	Federated Averaging (FedAvg, sample-size weighted)

volume, and the number of APOE $\varepsilon 4$ alleles, which gives $d = 8$ clinical features. The cohort has 3,700 subjects across all five ADNI phases: ADNI-1 (819), ADNI-GO (129), ADNI-2 (1,056), ADNI-3 (959), and ADNI-4 (737), counted by the phase in which each subject enrolled. Each subject is labelled with the diagnosis at the baseline visit, which gives $C = 3$ classes: cognitively normal (CN, 1,577 subjects), mild cognitive impairment (MCI, 1,566), and Alzheimer's disease dementia (AD, 557). MMSE and CDR-SB also enter the ADNI diagnostic criteria, so the clinical branch has partial access to label information through these two scores.

We have not yet obtained or processed the structural MRI scans for these subjects. For the evaluation, each subject's baseline T1-weighted scan will be skull-stripped, intensity-normalized, and resampled to a $32 \times 32 \times 32$ volume, and we will keep only subjects with both a scan and a complete clinical record. The data will be split by subject (70% training, 15% validation, 15% test), so no subject appears in more than one split.

B. Implementation parameters

The framework is implemented in PyTorch (v2.0) with Flower over gRPC [19], and the federated experiments will use $K = 5$ simulated clients. Table I lists the hyperparameters.

V. IMPLEMENTATION STATUS AND EVALUATION PROTOCOL

A. Implementation status

All components of Section III are implemented: the two encoders and the fusion classifier, FedAvg training over Flower with clipped and noise-perturbed client updates, and the Grad-CAM and SHAP explanation modules. The pipeline has run end to end on the real ADNI clinical variables, with synthetic volumes standing in for MRI scans. These runs confirm that the software executes, but we don't report their accuracy for two reasons. First, each synthetic volume was random noise whose intensity depended on the diagnosis label, so the image revealed the label, making any result involving the imaging branch meaningless. Second, the federated runs were scored on subjects that had also been used for training. The protocol below fixes both problems.

TABLE II
SYSTEM HYPERPARAMETERS AND EXECUTION SETTINGS

Parameter	Configuration
Image Backbone	3D ResNet-10 (MedicalNet, pretrained), 512-d Embedding
Image Input	80^3 Volume, Affine-Registered, Skull-Stripped, Brain Crop
Image Tabular Branch	MLP ($d \to 64$), Dropout 0.3 (Age, Sex, Region Features)
Image Head	FC ($576 \to 3$), Dropout 0.5
Image Optimization	AdamW ($\eta = 10^{-4}$ Backbone / 10^{-3} Head, Weight Decay 10^{-2})
Image Training	30 Epochs, Batch Size 8, Label Smoothing 0.1, Centralized
Clinical Net	MLP ($27 \to 32 \to C$), Dropout 0.1
Clinical Optimization	SGD ($\eta = 0.05$), Batch Size 16
Local Training	Local Epochs $E = 1$
Federation Bounds	Rounds $R = 10$ (ADNI) / 20 (Heterogeneity), Clients $K = 3$
Differential Privacy	L_2 Clip Norm $C = 2.0$, Noise Multiplier $\sigma = 1.8$, $\delta = 10^{-5}$
Data Skew Bounds	IID vs. Moderate / Strong / Quantity / Severe Non-IID
FL Framework	PyTorch + gRPC
Aggregator	FedAvg (Weighted); FedProx, SCAFFOLD, FedXAI-Adaptive Compared

TABLE III
PERFORMANCE ACROSS MODALITIES (HELD-OUT TEST PATIENTS)

Architecture Modality	Bal. Acc. (%)	Macro-F1	ROC-AUC
Clinical Vector Only (MLP, FedAvg + DP)	**83.4**	**0.83**	**0.950**
3D MRI Only (Logistic Regression)	42.6	0.40	0.599
3D MRI Only (Pretrained 3D ResNet-10)	45.6	0.42	0.653

CN / MCI / AD staging; macro ROC-AUC one-vs-rest. Clinical: 3 synthetic sites (900 / 600 / 300), $\varepsilon \leq 3.21$, 5 seeds. MRI: ADNI + OASIS, centralized, 215 test visits / 105 patients.

B. Planned comparisons

We will make four comparisons on real ADNI scans. (1) *Fusion:* under centralized training, we compare the fused model with an MRI-only 3D CNN and a clinical-only MLP. (2) *Federation:* we train the fused model with FedAvg on IID clients and on Dirichlet splits with $\alpha \in \{1.0, 0.5, 0.1\}$, and compare the results with centralized training. (3) *Update protection:* we run each federated setting with and without clipping and noise. (4) *Label leakage:* we also train every model with a clinical branch, excluding MMSE and CDR-SB.

C. Measures

We will score all models only on the held-out test subjects, which no client sees during training. The classes are unbalanced, so we will report accuracy, macro-averaged F1, one-vs-rest AUC, and per-class sensitivity, as mean and standard deviation over three random seeds. To compare convergence, we will record test accuracy at each communication round. For the explanations, we will compare the Grad-CAM maps with the brain regions known to be affected in AD, and compare the SHAP rankings of the clinical features across clients.

TABLE IV
FEDERATED DIAGNOSTIC PERFORMANCE UNDER DATA SKEW (FEDAVG AGGREGATION)

Data Distribution	ROC-AUC	PR-AUC	F1	Sens.	Spec.
Federated IID (600 / 600 / 600)	0.845	0.758	0.671	0.816	0.694
Moderate Non-IID	0.835	0.746	0.672	0.806	0.692
Strong Non-IID	0.812	0.711	0.676	0.771	0.699
Quantity Skew (1000 / 500 / 200)	0.829	0.730	0.667	0.775	0.730
Severe (Strong + Quantity)	0.826	0.667	0.594	0.776	0.692

MCI → AD progression, synthetic cohort, no DP, 20 rounds, mean of 10 seeds, test patients.

TABLE V
EXPLANATION CONSISTENCY ACROSS INSTITUTIONAL NODES (FEDAVG)

Stability Metric	Model	Value
SHAP Rank Correlation Between Sites (Spearman ρ)	Clinical MLP, no DP	0.915
SHAP Rank Correlation Between Sites (Spearman ρ)	Clinical MLP, with DP	0.911
SHAP Rank Correlation Between Runs (Spearman ρ)	Clinical MLP, no DP	0.506
SHAP Rank Correlation Between Runs (Spearman ρ)	Clinical MLP, with DP	0.233
3D Grad-CAM Region IoU (AD Regions, Top 10 %)	3D ResNet-10	0.025
Grad-CAM Top-10 % Voxels in AD Regions (Chance 0.015)	3D ResNet-10	0.027
Grad-CAM Map IoU Between Datasets	3D ResNet-10	0.415

SHAP: clinical MLP, synthetic sites, 5 seeds. Grad-CAM: 3D ResNet-10 (centralized, no DP), layer4, 911 test scans; ADNI / OASIS-2 / OASIS-1.

VI. CONCLUSION

We presented the design of FEDXAI, a federated multimodal framework for Alzheimer's disease diagnosis that keeps patient records at the hospital and explains its predictions. A 3D CNN and an MLP encoder are fused at the feature level and trained with FedAvg. Each client's update is clipped and perturbed with Gaussian noise, and every prediction comes with a 3D Grad-CAM heatmap and SHAP attributions. The software is implemented, and we have defined a protocol for evaluating it.

The work has real limitations. We have no accuracy results yet, because the framework hasn't been trained on real MRI scans. We claim no formal privacy guarantee: clipping is applied to each client's whole update, the noise level is low, and we do no privacy accounting. MMSE and CDR-SB are part of the ADNI diagnosis, so their contribution has to be measured by ablation. The 32^3 input limits Grad-CAM to coarse regions. And the clients will be simulated by splitting a single dataset.

Next we will run the evaluation on real ADNI scans. After that, we plan to add per-sample clipping with privacy accounting and secure aggregation, and to test on real multi-site data.

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